

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 July 2026
Team ID	
Project Name	Plugging into the future:An exploration of Electricity consumption of patterns
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Upload	Upload Consumption.csv file Import data from device sensors
FR-2	Data Visualization	Display daily electricity usage in bar chart Display monthly usage in line graph
FR-3	Device Analysis	Show power consumption by device: AC, Fridge, Fan, TV Highlight highest consuming device
FR-4	Bill Prediction	Calculate estimated monthly bill based on usage Show cost per device
FR-5	Energy Saving Suggestions	Suggest ways to reduce consumption Send alerts when usage is high

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard should be easy to understand for household users with simple charts and graphs
NFR-2	Security	User data from CSV file should be stored securely and no personal data should be exposed
NFR-3	Reliability	System should work consistently and show accurate electricity consumption data every time
NFR-4	Performance	Dashboard and charts should load within 3 seconds even with large CSV data
NFR-5	Availability	The web dashboard should be accessible 24/7 through browser
NFR-6	Scalability	System should handle data from multiple months and multiple devices without slowing down

